Actual norm matrices and bounded coordinate families

November 13, 2026. Unsigned producer evidence [pcc:unverified].

Actual matrix connection. For polynomial families A_i, B_i and a field-linear polynomial functional μ, define

$$E_{ik} = A_i A_k - f B_i B_k, \quad D_{ik} = \mu(E_{ik}), \qquad (C_j)_{ik} = [X^j]E_{ik} - [X^d]E_{ik}\,[X^j]G,$$

where $d = \operatorname{natDegree} G$. The actual moment matrix is symmetric. For $A_w = \sum_i w_i A_i$, $B_w = \sum_i w_i B_i$, $N_w = A_w^2 - f B_w^2$, Lean establishes

$$w^T D w = \mu(N_w), \quad w^T C_j w = [X^j]N_w - [X^d]N_w\,[X^j]G.$$

If G is monic, $\mu(G) = 0$ and the entry degrees are bounded by d, the preceding functional theorem gives the exact entrywise matrix identity

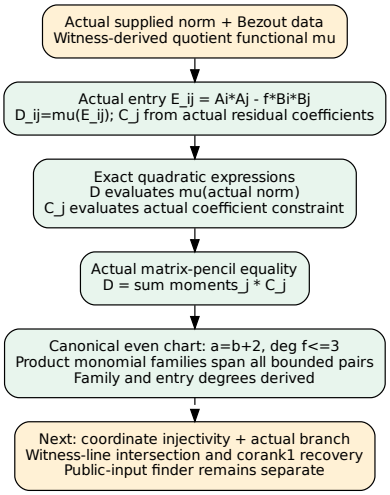
$$D = \sum_{j<d} \mu(X^j) C_j.$$

The final actual-witness theorem derives $\mu(G) = 0$ and the entry bounds; it does not assume matrix-pencil membership or numerical rank.

Canonical full chart. Supplied monic F, H, G satisfy

$$UF + V(fH) = 1, \quad (\alpha F)^2 - f(\beta H)^2 = cG, \quad c \neq 0.$$

Let $a = \deg F = b + 2$, $b = \deg H > 0$, $\deg G = 2a$, $\deg f \leq 3$. Use the actual quotient moment from the preceding theorem and the product monomial basis of the polynomial spaces of degrees $< a + 1$ and $< b + 1$. The first and second components supply the families. Their bounds and completeness are proved: every bounded pair (P, Q) is (A_w, B_w) for some common coordinate vector w. These canonical matrices satisfy the identity.



Exact scope, checks and next transfer

Formal source. `NormMomentMatrix.lean`, frozen source commit `cc5c9651a`. Ten declarations cover symmetry, actual norm/coefficient quadratics, matrix-pencil equality, family/entry degrees, bounded-pair completeness and the full even chart. Main declarations include `matrixPencil`, `familiesComplete` and `canonicalMatrixPencil`. The accompanying Markdown report maps every result to its exact source and hypotheses; ten producer bindings remain unsigned. No characteristic or separability restriction is introduced by this component.

Checks. Lean4.35.0 and unchanged pinned dependencies. Build006 passes3149 jobs; printed axioms are standard foundations only. Actual-source audit covers120 declarations with zero failing and zero additional trusted axioms. Separate fresh import-inclusive replay exits0; its own completion receipt is retained. No signed PCC verdict is claimed.

Build001 failed on intro applied to transpose equality and wrong function-sum simplification for Matrix. Matrix ext/sum/smul declarations resolve those errors. Build004 used unrestricted simp on projected basis reconstruction, reducing the supplied equation to True; controlled product/submodule sum rewrites resolve it. All failed and completed logs are retained. Earlier source hashes remain unchanged.

Supporting native record. `IDC1h345f3824264d232a` is copied unchanged from the preceding functional evidence. It checks the universal commutative-ring norm polarization at the previously recorded32+32 points, pinning that earlier exact source. No new native run is claimed. It does not check arbitrary matrix sums, pencil membership or rank; those components use their own Lean proofs. No Rust implementation changed in this round.

Next transfer. Prove coordinate injectivity and the arbitrary-pair bilinear evaluation. Define the actual target branch, exhibit a radical vector on which it is nonzero, and apply the earlier branch-test lemma. Compute the scaled witness-line intersection and transfer its dimension to exact matrix corank1; then apply guarded adjugate recovery with all nonzero/stationarity guards. Exceptional charts and a public-input moment finder remain separate. Current moments still use supplied witness data; the required $2^{o(n)}$ bit search cost does not follow from matrix existence.

Earlier completion proposal. These matrices supply a concrete rank-directed certificate form. Reachability through $W_{\mathrm{base}} + C(T)H(T)$ and an efficient choice of its free coefficients remain unestablished.

Sources and resources. Exact NormLift/NormDualPencil and functional source; pinned Matrix sum/smul, DegreeLT product basis, basis reconstruction, polynomial degree and product/submodule sum declarations. KB and conductor unavailable; bounded fallback, no priority claim. No paid job, empirical runtime/scaling campaign or budget override. Monthly isolab budget remains exhausted. No benchmark/leaderboard/ecdb changes. Source-linked Markdown, editable DOT/SVG, diagram PDF, this report and raw evidence accompany the result.